

Task.kt

```
@Entity(tableName = "tasks")
data class Task(
    @PrimaryKey(autoGenerate = true) val id: Int = 0,
    val title: String,
    val description: String,
    val isCompleted: Boolean = false
)
```

TaskDao.kt

```
@Dao
interface TaskDao {
    @Query("SELECT * FROM tasks")
    fun getAllTasks(): Flow<List<Task>>

    @Insert
    suspend fun insert(task: Task)

    @Delete
    suspend fun delete(task: Task)
}
```

AppDatabase.kt

```
@Database(entities = [Task::class], version = 1)
abstract class AppDatabase : RoomDatabase() {
    abstract fun taskDao(): TaskDao
}
```

TaskRepository.kt

```
class TaskRepository(private val dao: TaskDao) {
    val allTasks: Flow<List<Task>> = dao.getAllTasks()
    suspend fun insert(task: Task) = dao.insert(task)
    suspend fun delete(task: Task) = dao.delete(task)
}
```

TaskViewModel.kt

```
class TaskViewModel(application: Application) : AndroidViewModel(application) {
    private val db = Room.databaseBuilder(
```

```

        application, AppDatabase::class.java, "task-db"
    ).build()
    private val repo = TaskRepository(db.taskDao())

    val allTasks = repo.allTasks.asLiveData()

    fun addTask(title: String, desc: String) {
        viewModelScope.launch {
            repo.insert(Task(title = title, description = desc))
        }
    }
}

```

AddTaskScreen.kt

```

@Composable
fun AddTaskScreen(onAdd: (String, String) -> Unit, onCancel: () -> Unit) {
    var title by remember { mutableStateOf("") }
    var desc by remember { mutableStateOf("") }

    Column(Modifier.padding(16.dp)) {
        TextField(value = title, onValueChange = { title = it }, label = {
            Text("Titre") })
        TextField(value = desc, onValueChange = { desc = it }, label = {
            Text("Description") })
        Row(Modifier.fillMaxWidth(), horizontalArrangement =
            Arrangement.SpaceEvenly) {
            Button(onClick = onCancel) { Text("Annuler") }
            Button(onClick = { onAdd(title, desc) }) { Text("OK") }
        }
    }
}

```

MainActivity.kt

```

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContent {
            val navController = rememberNavController()
            val viewModel: TaskViewModel = viewModel()

            NavHost(navController, startDestination = "list") {
                composable("list") {

```



```

        Text(task.description)
    }
}
}
Button(onClick = onAdd, modifier = Modifier.fillMaxWidth()) {
    Text("Ajouter une tâche")
}
}
}

```

DetailActivity.kt

```

class DetailActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        val title = intent.getStringExtra("title") ?: ""
        val desc = intent.getStringExtra("desc") ?: ""

        setContent {
            Column(Modifier.padding(16.dp)) {
                Text("Titre : $title", fontWeight = FontWeight.Bold)
                Text("Description : $desc")
                Button(onClick = { finish() }) { Text("Retour") }
            }
        }
    }
}

```